ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Area & Volume of Similar Shapes

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

AREA & VOLUME OF SIMILAR SHAPES · CH4 · L9

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The Map

This lesson collapses Area & Volume of Similar Shapes into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Find the length scale factor

TRIGGER *"work out", "find", a missing value*

02 Square it for area

TRIGGER *"similar", "scale factor", "area/volume"*

03 Cube it for volume

TRIGGER *"similar", "scale factor", "area/volume"*

04 Reverse from area or volume

TRIGGER *"similar", "scale factor", "area/volume"*

05 Split the ratio

TRIGGER *"similar", "scale factor", "area/volume"*

BANK EVIDENCE 36 website questions, 36 checked solutions, 3 local PDFs read.

01 Find the length scale factor

TRIGGER *"work out", "find", a missing value*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$15 : 45 = 1 : 3$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2023 P1H Q23

The small and large square
This is a reasoning step: write one clear sentence, then continue the calculation.

Their heights are

$$15 : 45 = 1 : 3$$

Finish: $\boxed{5090 \text{ cm}^2}$

FORMULA BANK

Length scale factor k. Area scale factor k^2. Volume scale factor k^3.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram. Diagram NOT accurately drawn 15 cm 39 cm 39 cm 45 cm The height of the small pyramid is 15 cm. The height of the large pyramid is 45 cm. The square base of the large pyramid has side length 39 cm. Work out the total surface area of the frustum. Give your answer correct to the nearest whole number. cm 2

YOUR SOLUTION

02 Square it for area

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$15 : 45 = 1 : 3$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
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WORKED MODEL

Source pattern: May 2023 P1H Q23

The small and large square
This is a reasoning step: write one clear sentence, then continue the calculation.

Their heights are

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FORMULA BANK

Length scale factor k. Area scale factor k^2. Volume scale factor k^3.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram. Diagram NOT accurately drawn 15 cm 39 cm 39 cm 45 cm The height of the small pyramid is 15 cm. The height of the large pyramid is 45 cm. The square base of the large pyramid has side length 39 cm. Work out the total surface area of the frustum. Give your answer correct to the nearest whole number. cm 2

YOUR SOLUTION

03 Cube it for volume

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{h}{2h} = \frac{1}{2}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

> **Source pattern:** Jan 2020 P1HR Q20
>
> **The two cones are similar**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **The large cone has height**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **Finish:** $r = \frac{7h}{16}$

FORMULA BANK

Length scale factor k. Area scale factor k^2. Volume scale factor k^3.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 20 The diagram shows a frustum of a cone and a sphere. The frustum is made by removing a small cone from a large cone. The cones are similar. h cm 2h cm r cm r cm Diagram NOT accurately drawn The height of the small cone is h cm. The height of the large cone is 2h cm. The radius of the base of the large cone is r cm. The radius of the sphere is r cm. Given that the volume of the frustum is equal to the volume of the sphere, find an expression for r in terms of h. Give your expression in its simplest form. r =

YOUR SOLUTION

04 Reverse from area or volume

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{h}{2h} = \frac{1}{2}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q20

The two cones are similar

This is a reasoning step: write one clear sentence, then continue the calculation.

The large cone has height

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $r = \frac{7h}{16}$

FORMULA BANK

Length scale factor k. Area scale factor k^2. Volume scale factor k^3.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 20 The diagram shows a frustum of a cone and a sphere. The frustum is made by removing a small cone from a large cone. The cones are similar. h cm 2h cm r cm r cm Diagram NOT accurately drawn The height of the small cone is h cm. The height of the large cone is 2h cm. The radius of the base of the large cone is r cm. The radius of the sphere is r cm. Given that the volume of the frustum is equal to the volume of the sphere, find an expression for r in terms of h. Give your expression in its simplest form. r =

YOUR SOLUTION

05 Split the ratio

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$9 : 13$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

> **Source pattern:** Jan 2020 P2HR Q20
>
> **The two vases are similar**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **Their heights are in the**
>
> $$9 : 13$$
>
> **Finish:** $\boxed{583.2\ \text{cm}^2}$

FORMULA BANK

Length scale factor k. Area scale factor k^2. Volume scale factor k^3.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 20 The diagram shows two similar vases, A and B. Diagram NOT accurately drawn 9 cm 13 cm A B The height of vase A is 9 cm and the height of vase B is 13 cm. Given that surface area of vase A + surface area of vase B = 1800 cm2 calculate the surface area of vase A. cm2

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "find", a missing value	Find the length scale factor
"similar", "scale factor", "area/volume"	Square it for area
"similar", "scale factor", "area/volume"	Cube it for volume
"similar", "scale factor", "area/volume"	Reverse from area or volume
"similar", "scale factor", "area/volume"	Split the ratio

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.